Sahil Mohril

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B.Tech Computer Science (Data Science) student with a strong foundation in Java Spring backend architecture, data science methodologies, and interactive visualization. Adept at leveraging core software engineering principles alongside machine learning frameworks to extract actionable insights and solve complex, multi-disciplinary problems. Committed to continuous optimization, proactive skill growth, and delivering robust software solutions within collaborative environments.

Education

Vellore Institute of Technology **CGPA: 9.15** Vellore, India
B.Tech in Computer Science and Engineering (Data Science) *2023 – 2027*

Seth M.R. Jaipuria School Score: 93.5% Lucknow, India
Higher Secondary School (ISC) *2022*

Seth M.R. Jaipuria School Score: 96.0% Lucknow, India
Secondary School (ICSE) *2020*

Technical Skills

Languages: Java, Python, C, C++, SQL, R
Web & Backend: Spring Boot, React, HTML, REST APIs
Databases: MySQL, PostgreSQL, MongoDB
Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Exploratory Data Analysis
Big Data & Cloud: Apache Spark (PySpark), Apache Kafka, AWS
Developer Tools: Git, GitHub, Databricks, Tableau, VS Code

Projects

Predictive Maintenance System for Rotating Machinery | *Python, Scikit-learn, Signal Processing* **[GitHub]**
- Built an end-to-end predictive maintenance pipeline combining vibration signal processing with machine learning to estimate Remaining Useful Life (RUL) of bearings.
- Engineered time-domain and frequency-domain features from raw sensor signals and trained regression models to predict degradation trends.
- Validated pipeline on benchmark bearing run-to-failure datasets, improving early fault-detection accuracy over baseline threshold methods.

MapReduce Engine in Java | *Java, Multithreading, Distributed Systems* **[GitHub]**
- Designed and implemented a distributed MapReduce framework from scratch in Java, supporting custom Mapper and Reducer interfaces similar to Hadoop's programming model.
- Built a master-worker architecture to partition input data, schedule map and reduce tasks across worker threads/nodes, and shuffle intermediate key-value pairs.
- Implemented fault tolerance with task re-execution on worker failure and used the engine to run word-count and large-scale aggregation, demonstrating linear speedup with added workers.

TimberTrail Log Analytics | *PySpark, Python, Streamlit* **[GitHub]**
- Built a scalable log analytics pipeline ingesting raw HTTP server log entries to parse unstructured Combined Log Format text into structured columns with a 0% parse-failure rate.
- Engineered a DDoS detection system by aggregating error responses per IP successfully flagging malicious IPs generating 700+ error requests across 404/500 status codes.
- Developed an interactive Streamlit BI dashboard reading directly from the Parquet store to visualize traffic KPIs, hourly trends, and bad actors, replacing manual log grep workflows with a queryable data layer.

Certifications

Oracle Certified Java Professional (SE 11) | *Oracle*

Data Science Course, Cognizance '24 | *IIT Roorkee*

NumPy and Pandas Masterclass | *Udemy*

MySQL Bootcamp | *Udemy*